USB POWER S9.2, +/−12V ONLY, 15 V PD INPUT (ROUTING−READY, NOT ROUTED)

Own−input enable: OVP−coordinated VIN clamp: PGND star at C82: Kelvin R109: STARTED one−shot: no crowbar.

USB PD, main eFuse and VIN damper



File: 01_usb_input.kicad_sch

+12 V buck−boost converter



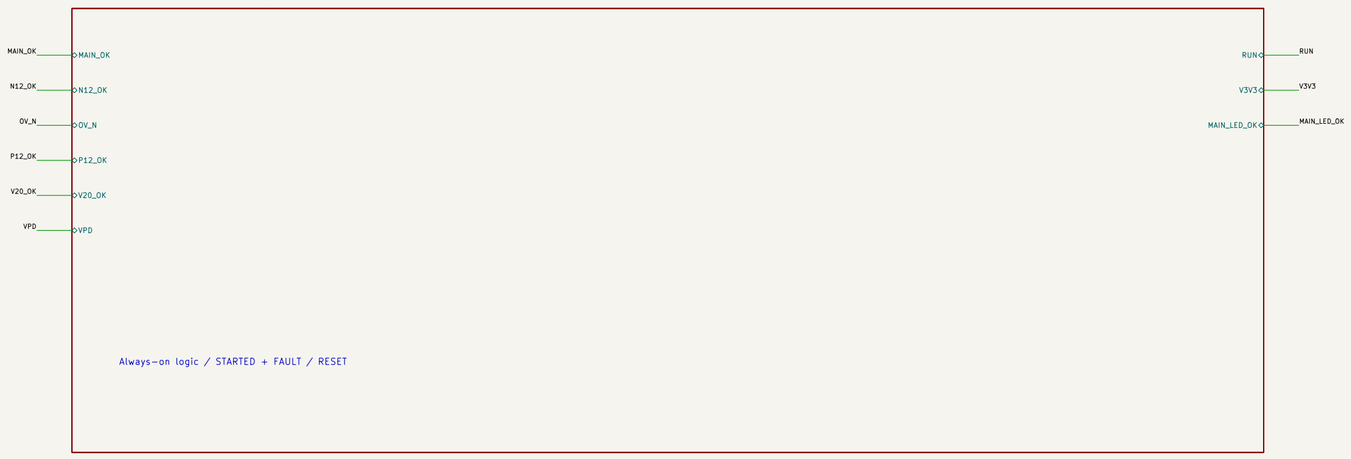
File: 02_positive_12v.kicad_sch

−12 V inverting converter



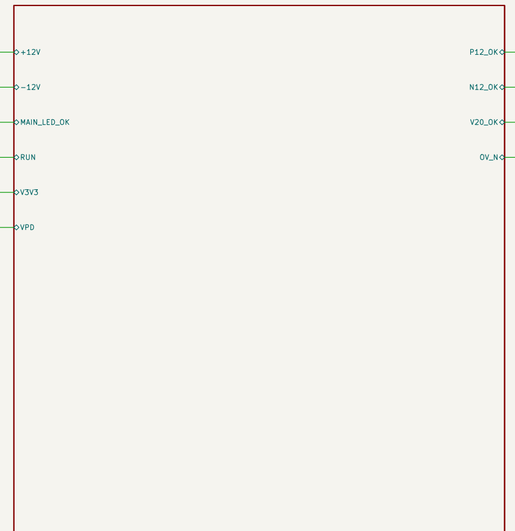
File: 03_negative_converter.kicad_sch

Always−on control − fault latch, overvoltage preset and RESET



File: 04_fault_reset.kicad_sch

Two output headers − monitors, overvoltage guard and MAIN LED



File: 05_outputs_status.kicad_sch

ONE−SHOT FAULT POLICY
STARTED retains the first start within each power epoch.
Faults cannot restart it: full power reset CAN. Repeated source resets FAIL qualification.
OV_N presets FAULT asynchronously. Crowbar vs eFuse deferred/open.

+12V 2.4 A / −12V 0.8 A, <=33 W together (prototype, not measured; thermal rating conditional on the routing contract). 13.5−16.5 V at module; 15 V / 3 A PD (start into the full rack needs ~4.65 A for ~2.5 ms: acceptance test S1). Rack <=0.44 mF. No reverse−current blocking: no powered parallel supply or hotplug.

One−shot arming latch (STARTED): U18/U20 on VPD: see docs/design.md
S9.3−10P: +/−12V, 15 V PD; MAIN LED only; two 10−pin outputs, routing in progress

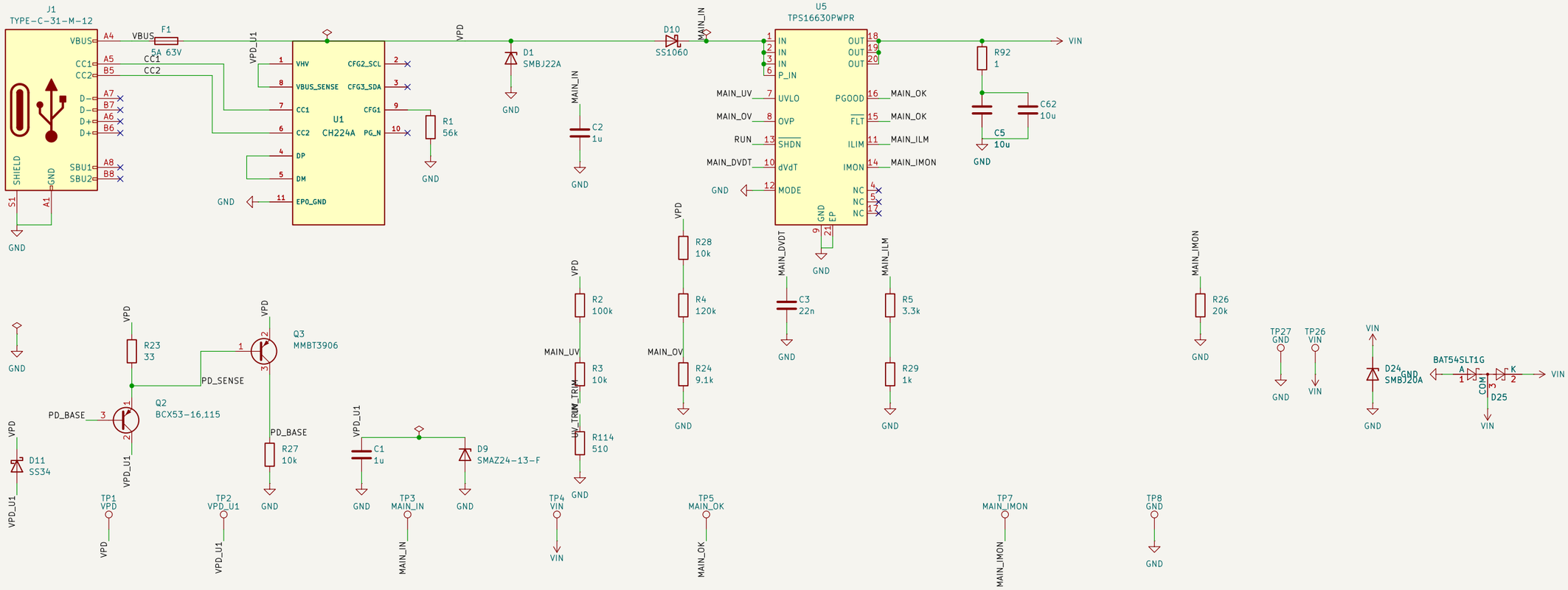
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Sheet: /
File: usb−power−45w.kicad_sch

Title: USB power S9 board − wired power flow overview

Size: User	Date:	**Rev: S9.3−10P**
KiCad E.D.A. 10.0.6		Id: 1/6

USB PD, MAIN EFUSE AND VIN DAMPER



S9.2: 15 V / 3 A PD, R1 56k, U5 UVLO: R2 100k 0.1% / (R3 10k 0.1% + R114 510R 1%), worst start <13.0 V.
U5 OVP R28+R4 130k / R24 9.1k: rises 17.30–19.44 V, recovers 16.03–18.27 V (S9.1: 6.8k, 24.1 V).
D24 SMBJ20A: VR 20 V > OVP max, VBR 22.2 V (21.7 V at 0 C), 32.4 V at 18.6 A < U2 42 V: OVP acts first.
D25: low-current negative clamp, slew-limited room-temperature test only. TP26/TP27 = D24 clamp probe pair.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇— MAIN_OK

RUN ◇— RUN

VIN ◇— VIN

VPD ◇— VPD

S9.2 routing-ready; P2.2 UNROUTED; docs/design.md

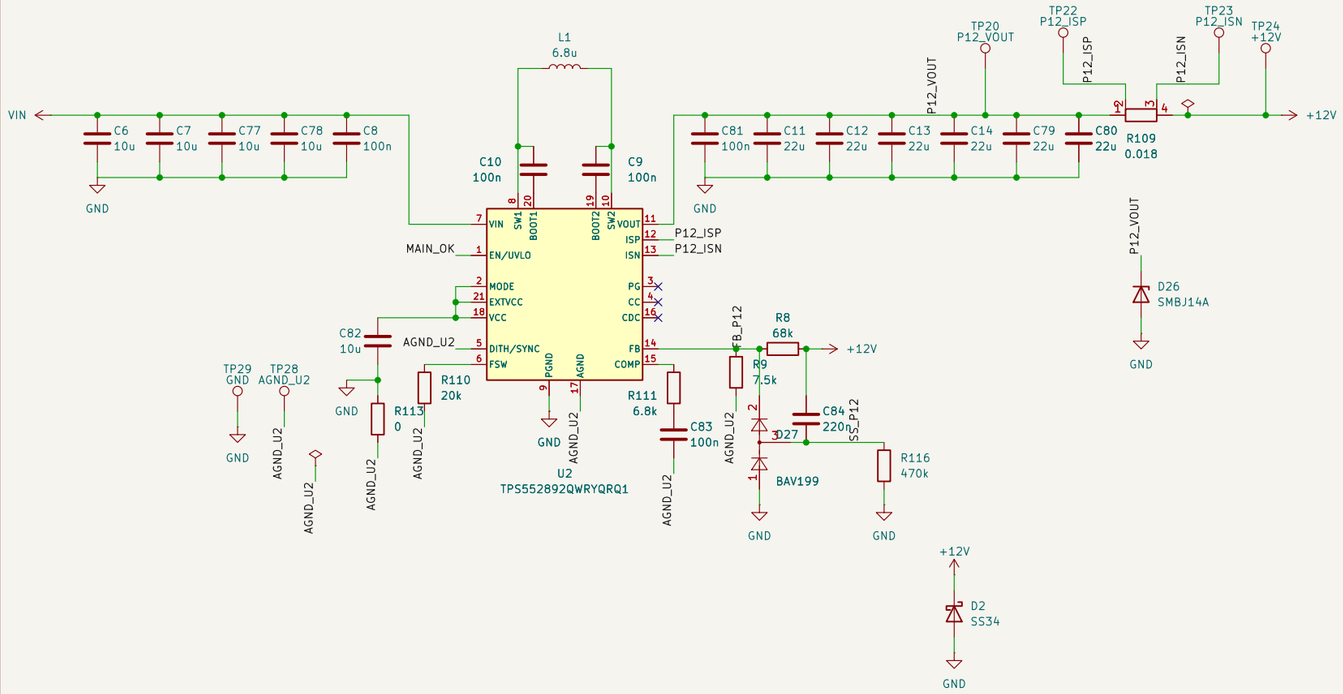
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S9.2**
KiCad E.D.A. 10.0.6		Id: 2/6

+12 V BUCK–BOOST CONVERTER



S9.2: TPS552892-Q1, 966 kHz, L1 6.8u; R109 18mR 1W on a 4-terminal Kelvin land (2.62–2.95 A).
U2 ISP/ISN on their own sense nets P12_ISP/P12_ISN (R109 pads 2/3 only). C82 ground terminal on PGND.
AGND_U2: U2.5/.17, R9, R110, C83, TP28. ONLY join to PGND through R113 at C82.2 (3–4 PGND vias there).
COMP R111 6.8k / C83 100n (zero 234 Hz). Rack <=0.44 mF. Acceptance pads TP20/22/23/24/28/29.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

S9.2 routing-ready; P2.2 UNROUTED; docs/design.md

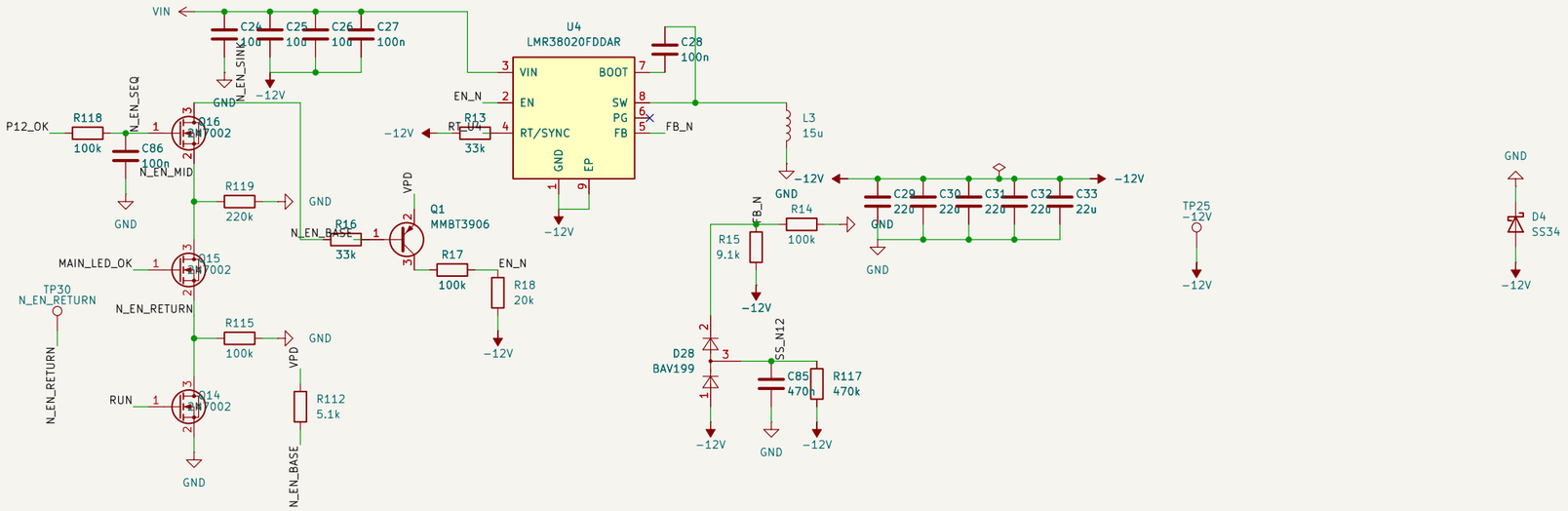
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Sheet: /+12 V buck–boost converter/
File: 02_positive_12v.kicad_sch

Title: +12 V buck–boost converter

Size: User	Date:	**Rev: S9.2**
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−12 V INVERTING CONVERTER



S9.1/S9.2: U4 enable is powered from own VPD upstream of D10, NEVER from +12V or pumped VIN.
Q14 RUN and Q15 MAIN_LED_OK form an AND sink for Q1 base; R112 returns base to emitter when either is off.
S9.2: R115 100k defines N_EN_RETURN (Q15 then only a source follower: Q1 VEB <= 0.18 V). TP30 probes it.
Unpowered or latched-off live +/−12V bus cannot self-enable U4 (truth table derived from this netlist).
LMR38020F power stage unchanged. Powered parallel supplies and unrestricted hotplug remain prohibited.

INTER-SHEET INTERFACE (see wired power-flow overview)

−12V ◇ −12V

MAIN_LED_OK ◇ MAIN_LED_OK

P12_OK ◇ P12_OK

RUN ◇ RUN

VIN ◇ VIN

VPD ◇ VPD

S9.2 routing-ready; P2.2 UNROUTED; docs/design.md

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Sheet: /−12 V inverting converter/
File: 03_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S9.2**
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ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET

U9
TPS7A1633DGNR
C46 1u
C47 10u
R63 10k
MAIN_OK

R64 10k
RUN

U10
TPS3808G01DBVR
R91 15k
R65 47k
R66 10k
POR_SEN
R67 10k
R68 10k
C48 100n

U11
TPS3808G01DBVR
R69 33k
R70 10k
TIMER_GO
START_SENSE
C49 100n
START_CT
R71 10k
ARMED
C50 100n

SW1
TL3340AF160QG
RESET_BTN_N

U12
SN74HC74DR
C52 100n

U13
SN74HC08DR
C53 100n

U14
SN74HC20DR
C54 100n

R72 10k
C51 100n
ANY_BAD
BAD_RC

U15
SN74LVC1G17DBVR
C55 100n

TP9 RUN
TP10 ARMED
TP11 FAULT_LATCH
TP12 P12_OK
TP14 N12_OK
TP15 V20_OK
TP16 V3V3
TP6 OV_N
TP19 STARTED

S9.1 retains S8.5 STARTED one-shot: START_N = NAND(MAIN_LED_OK,RUN); TIMER_GO = RUN AND STARTED.
U11 -> ARMED after 0.31–0.88 s. RESET clears STARTED/FAULT; U20 OV_N presets FAULT even before ARMED.
A full V3V3/POR reset DOES re-arm; qualification rejects repetitive hard-reset sources; no masking by retries.
ANY_BAD = NAND(MAIN_LED_OK,P12_OK,N12_OK,V3V3). R83 100k validity floor unchanged.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK
MAIN_OK
N12_OK
OV_N
P12_OK
RUN
V20_OK
V3V3
VPD

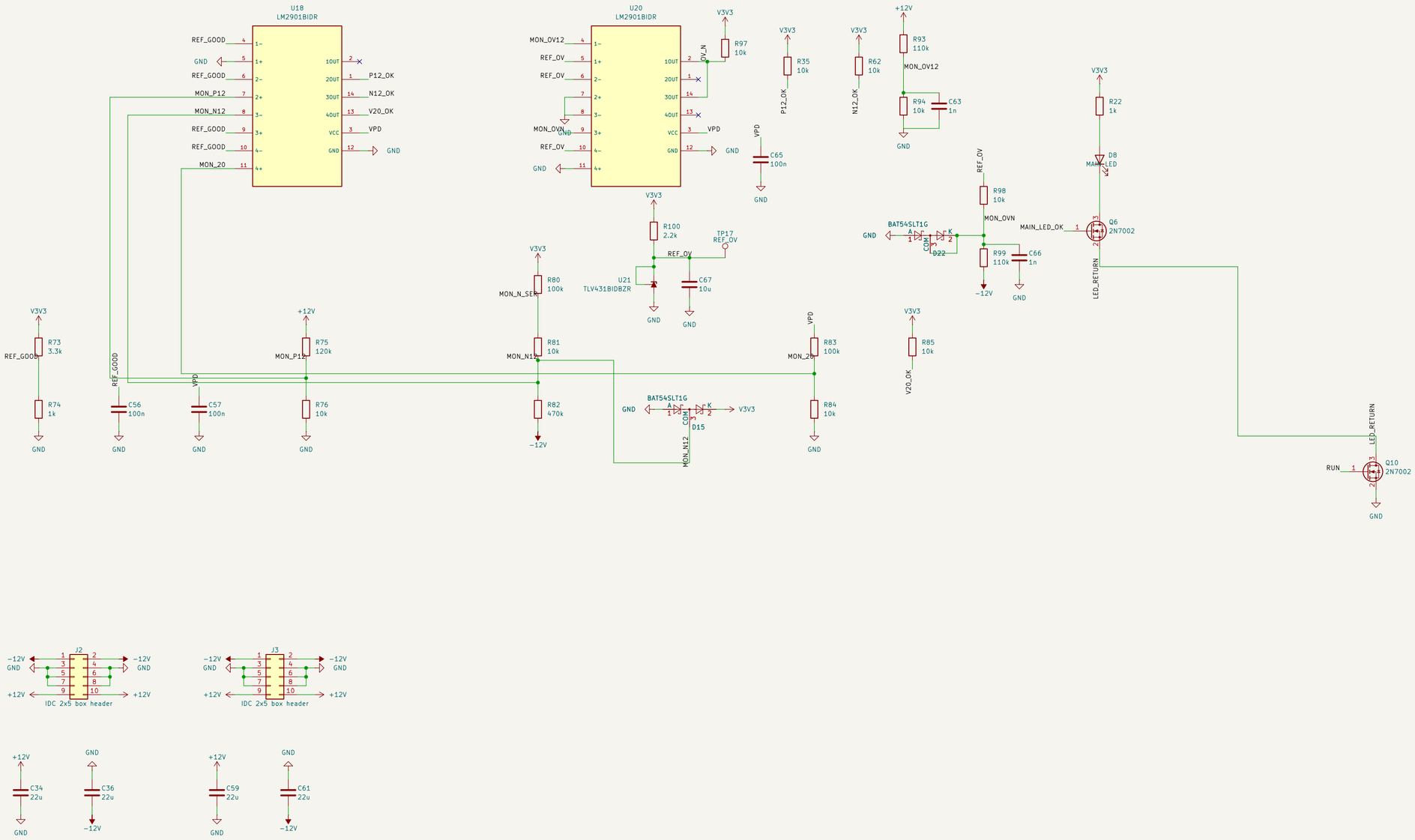
S9.2 routing-ready; P2.2 UNROUTED; docs/design.md
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Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 04_fault_reset.kicad_sch

Title: Always-on control — fault latch, overvoltage preset and RESET

Size: User	Date:	**Rev: S9.2**
KiCad E.D.A. 10.0.6		Id: 5/6

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

S9.3-10P: +/-12V only, J2/J3 10-pin IDC (1/2 -12V, 3-8 GND, 9/10 +12V). MAIN LED only. U18/U20 on VPD, own REF_OV unchanged.
Guard +12V 14.48–15.27 V, −12V magnitude 13.27–14.02 V. Guard disconnects INPUT, not an external bus.
No rail crowbar or eFuse. Uninjected rails decay with load; injected rail remains at the external voltage.
See docs/design.md and docs/design/routing-notes.md for acceptance tests and routing rules.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V · −12V · MAIN_LED_OK · N12_OK · OV_N

P12_OK · RUN · V20_OK · V3V3 · VPD

S9.2 routing-ready; P2.2 UNROUTED; docs/design.md

Big Modular

Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 05_outputs_status.kicad_sch

Title: Two output headers — monitors, overvoltage guard and MAIN LED

Size: User	Date:	Rev: S9.2
KiCad E.D.A. 10.0.6		Id: 6/6